WPRS — Whisper Position Reporting System

Infrastructure-free position beaconing over HF with a 200 mW WSPR transmitter and a car-roof 14 MHz antenna: design and three seasons of measurements

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WSPR Logger project note — version 1.1 — 6 November 2026 — `github.com/oh2gax/wspr_logger`

Abstract

WSPR, the Weak Signal Propagation Reporter protocol, carries a callsign, a Maidenhead locator and a power level in a 6 Hz-wide, two-minute transmission that is decoded by volunteer receivers worldwide and published in an open database. If the locator is computed from a GPS fix and the transmitter is moved, the stream of reports becomes a position track that needs no mobile network, no VHF repeater or APRS coverage and no receiver of one's own. This note describes that idea, here called the *Whisper Position Reporting System* (WPRS), and evaluates it with a ZachTek WSPR desktop transmitter (about 200 mW, built-in GPS) driving a Diamond monoband 14 MHz whip on a magnetic mount on the roof of a car. The evaluation uses 42 723 logged transmissions from 2024-07-04 to 2026-11-06, spread over three summer seasons. A six-character locator fixes the position to a square of 4.6×4.6 km at 60°N (r.m.s. error 1.9 km). The transmitter sends every ten minutes; consecutive fixes are exactly ten minutes apart in 91 % of the cases, the probability of having a fix from the previous hour is 92 % in the daytime (08–16 UTC) and 38 % at 01–04 UTC, and the median fix is confirmed by 17 independent receivers (85 % by three or more). The system tracked a local commute at one-square resolution and recorded trips to 193 km from home. A 30 000 mAh power bank, predicted to last 6.2 days at the measured duty cycle, lasts four to five days in practice. Reducing the output to 25 mW would have kept 89 % of the receiving stations; to 10 mW, 70 %.

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1 Motivation

Position tracking of a vehicle normally relies on one of two infrastructures: a cellular network (a GPS tracker that reports over mobile data) or an amateur-radio VHF network such as APRS, which needs a digipeater or an internet gateway within range of the vehicle. Both are absent in much of the countryside, on the sea, in the mountains and abroad. The HF bands are different: the ionosphere returns HF signals to the ground so that a transmission of a fraction of a watt can be heard thousands of kilometres away, and the WSPR network is a worldwide set of volunteer receivers [1] (3 713 different stations heard this beacon during the three seasons analysed here). If the transmitter simply states where it is, *somebody* will hear it, and the public database holds the answer for anyone who looks.

This is the idea of WPRS. It is deliberately modest: a position is delivered to a square of a few kilometres every ten minutes, with no acknowledgement, over a path that is sometimes not available. The aim of this note is to describe the technique, to quantify what it achieves from an ordinary car with a small whip antenna and 200 mW, and to say honestly where it is useful and where it is not. The measurements come from the author's own station, with the WSPR Logger software [5] recording every transmission and every reception.

2 Concept and protocol background

2.1 The WSPR signal

A WSPR transmission lasts about 110.6 s, starts one second after an even UTC minute, and consists of 162 symbols of four-tone frequency-shift keying at 1.4648 baud with a tone spacing of the same size; the occupied bandwidth is about 6 Hz [3]. Transmissions are made in a 200 Hz wide segment of each band; on 20 m this is 14 097 000–14 097 200 Hz (dial frequency 14.0956 MHz plus an audio offset of 1 400–1 600 Hz). The ZachTek transmitter picks a random frequency within this segment for each transmission, which spreads the transmitters over the segment [4]. The receiver decodes the symbols with error correction down to a signal-to-noise ratio of around −30 dB in a 2 500 Hz reference bandwidth; in the data of this note the lowest reported value is -35 dB and the weakest decodes cluster near -34 dB.

2.2 Message types and the six-character locator

A WSPR message carries 50 information bits. The three message types that matter here are

Type 1
callsign (up to six characters), four-character Maidenhead locator, power in dBm;

Type 2
callsign with a prefix or suffix (a "compound" callsign) and power, without a locator;

Type 3
a 15-bit *hash* of the callsign, a six-character locator and the power.

A six-character locator therefore needs a Type 3 message, in which the callsign is represented only by its hash; a receiver that has also decoded a Type 1 message from the same station can resolve the hash back to the callsign. A beacon that wants to send a precise position thus normally alternates the message types (or sends Type 3 occasionally). Which scheme a particular transmitter uses is a firmware and configuration matter described in the manufacturer's documentation [4]; what matters for this note is the result: every one of the 42 723 transmissions logged here has a six-character locator in the public database, and the locator follows the car.

2.3 Resolution of a Maidenhead locator

The Maidenhead locator divides the Earth into 18×18 fields of 20° by 10°, each into 10×10 squares of 2° by 1°, and each of these into 24×24 subsquares of $5'$ by $2.5'$ (Table 1). At the home latitude of 60.3°N a subsquare measures 4.6 km north–south by 4.6 km east–west, an area of 21.3 km^2. If the true position is uniformly distributed in the square and the centre is quoted, the root-mean-square position error is

$$\sigma = \sqrt{\frac{a^2 + b^2}{12}} = 1.9 \text{ km}, \tag{1}$$

with a and b the side lengths, and the largest possible error is half the diagonal, 3.3 km. The four-character locator of a Type 1 message is much coarser: 111 km by 110 km at this latitude. A six-character locator carries $\log_2(18^2 \cdot 10^2 \cdot 24^2) = 24.2$ bits (18 662 400 distinct squares) against 15.0 bits for four characters.

Table 1. Maidenhead locator levels. Size in kilometres is for 60.3°N.

Level	Example	Size (degrees)	Size (km)
Field	KP	20×10	$\approx 1\,100 \times 1\,110$
Square	KP20	2×1	$\approx 110 \times 111$
Subsquare	KP20jg	$5' \times 2.5'$	4.6×4.6

2.4 The path of a fix

A WPRS fix travels from the transmitter through the ionosphere to any receiver that happens to be listening, from there over the internet to the WSPR database [1], and from the database (here via `wspr.live` [2]) to whoever looks at the track. The delay is not controlled by the transmitter: receivers upload after the two-minute decode, distant stations tend to upload later, and the logger polls every ten minutes, so a fix is typically visible about six to ten minutes after its start. WPRS is therefore a *breadcrumb* system, not a live tracker.

3 Station

3.1 Transmitter

The transmitter is a ZachTek WSPR desktop transmitter [4]. According to the manufacturer it generates the signal with a Si5351 PLL clocked from a temperature-compensated crystal oscillator, amplifies the square wave in a line-driver IC to about 23 dBm (200 mW) into 50 Ω on most bands and removes the harmonics with four switched low-pass filters plus a VHF filter (overtones attenuated by 50 dB or more). A built-in GPS module and an external active GPS antenna (3 m cable, magnetic base) provide time and position; the firmware, running on an Arduino Pro Mini, encodes the WSPR message. It is powered from 5 V USB and draws about 100 mA at idle and 250 mA while transmitting. A Windows configuration program sets callsign, power, bands and start mode; once configured, the unit runs without a PC. Three status LED patterns are useful in the car: single blinking means it is acquiring the position, double blinking that it is waiting for the start of an even minute, and fast periodic blinking that it is transmitting.

3.2 Antenna and installation

The antenna is a Diamond monoband 14 MHz whip on a magnetic mount at the roof of a VW Touran; the car body is the ground plane. A magnetic mount on a roof is a practical compromise: it is installed and removed in seconds, but its performance depends on the contact area and on the capacitive coupling to the roof, and a physically short 20 m whip has a much lower efficiency than a full-size quarter-wave vertical. The SWR and the radiation pattern were not measured for this note, so the effective radiated power is unknown; the receiver reports in Section 5 are the only measure of what actually leaves the car. The GPS puck sits where it sees the sky. In winter the antenna is not fitted, because of snow and low temperatures, and the transmitter is not used in the car; the dataset therefore covers March to November only.

3.3 Power supply and energy budget

The transmitter is powered from a 30 000 mAh USB power bank. Two packs are rotated: when one is empty (after four to five days) the other, fully charged, replaces it and the empty one is recharged. With a nominal cell voltage of 3.7 V the pack stores 111 Wh; assuming 85 % conversion efficiency to 5 V this leaves 94 Wh. A WSPR transmission occupies 110.6 s of every 600 s (18.4 %), so the average load is 128 mA, or 0.64 W (15.3 Wh per day), and the predicted runtime is 6.2 days (Table 2). The observed four to five days correspond to 65–81 % of the prediction, which is plausible given the cold nights, the real conversion efficiency of power banks at small load, and the GPS puck current that is not counted here.

Table 2. Energy budget of the 30 000 mAh battery pack (nominal 3.7 V, 111 Wh; 85 % assumed conversion efficiency to 5 V).

Load case	Current (mA)	Power (W)	Runtime (days)
Idle (GPS acquiring / waiting)	100	0.50	7.9
Transmitting continuously	250	1.25	3.1
WPRS duty cycle (18.4 % transmitting)	128	0.64	6.2

4 Data and methods

The logger polls the public WSPR database every ten minutes and stores, for the beacon's callsign, one record per heard transmission (timestamp, six-character locator and its centre coordinates, number of reporting stations, largest distance); the *replay database* also stores each reporting station per poll. The earlier seasons were completed afterwards from `wspr.live` by the script `tools/backfill.py`, which downloads every reception of the callsign and rebuilds the transmission records and the poll snapshots; a record built this way could contain late reports that the live poll missed. A check on the live-logged days 1 June 2026 to 7 June 2026 (995 snapshots rebuilt and compared with the stored ones) found 41.0 stations per snapshot stored against 41.0 rebuilt (ratio 1.001), so the effect is negligible; a fix needs only one report, which makes the fix counts and positions even less sensitive to it. This note uses 42 723 transmissions on 490 of the 856 calendar days from 2024-07-04 to 2026-11-06 (the car is not used in winter), and the 3 092 119 station reports of the replay database. A *fix* is a transmission that was heard by at least one station.

Availability at time t is the probability that the most recent fix is no older than τ ($\tau = 0$: the current slot, 30 and 60 minutes), evaluated on every transmit slot of the 474 days with at least 20 fixes and averaged per UTC hour. A missing fix means either that no receiver heard the transmission or that the transmitter was off; the logger cannot distinguish the two. **Trips** are runs of fixes more than 25 km from the home square, merged when separated by less than eight hours. **Power scaling.** Lowering the output by $\Delta = 10 \log_{10}(200\,\mathrm{mW}/P)\,\mathrm{dB}$ lowers every received SNR by the same amount (the receiver noise is unchanged); a station is counted as *still decoding* if the best SNR it ever reported minus Δ stays at or above the decoding floor of -34 dB. This is an upper bound for the sustained reach, because a station's best SNR is an extreme value. Reports from one station whose distance contradicted its own grid were removed (see the companion study).

5 Results

5.1 Cadence and availability of fixes

All logged transmissions start on minutes ending in 2, i.e. one transmission every ten minutes, so at most 144 per day. On the median day 91 transmissions were heard (maximum 143), 63 % of the possible ones (Figure 1a). Between two consecutive fixes the interval is exactly ten minutes in 91 % of the cases, at most 30 minutes in 97 % and at most 60 minutes in 98 %; 242 gaps exceed six hours (nights without reports, periods when the transmitter was off and the winter breaks).

The availability depends strongly on the time of day (Figure 1b), because it follows the opening of the band. A fix exists for the current slot in 85 % of cases between 08 and 16 UTC and in 22 % between 01 and 04 UTC. The probability of having a fix from the previous 60 minutes averages 75 % over the day, 92 % in the daytime, 38 % at night and has its minimum of 31 % at 2 UTC. A WPRS user must plan for this: during the day the position is almost always less than an hour old, at night it often is not.

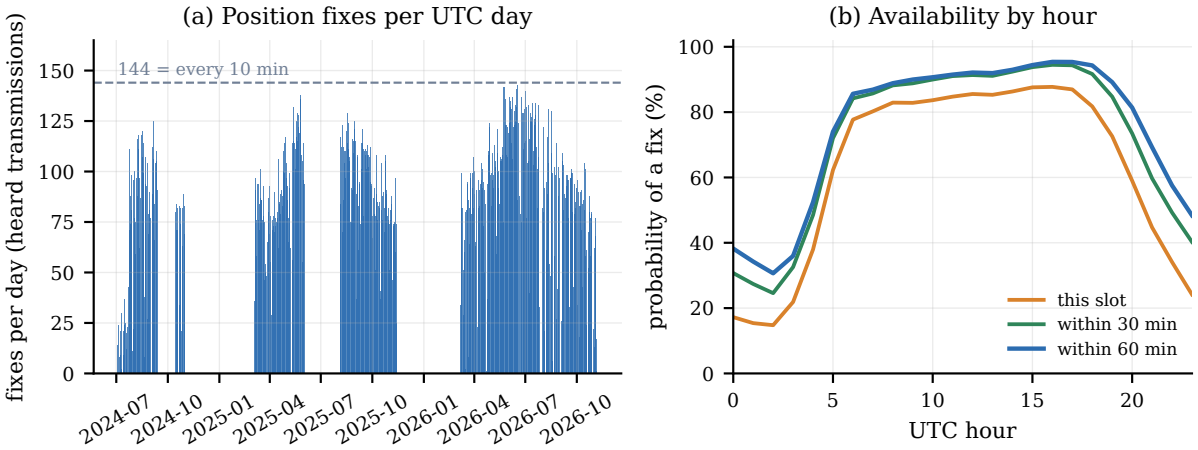


Figure 1. (a) Position fixes per UTC day. (b) Probability of a fix by UTC hour: from the current slot, from the previous 30 minutes and from the previous 60 minutes.

5.2 Confirmation by several receivers

A fix heard by one station only is fragile: a wrong grid in the receiver's own report or a decoding error produces a wrong position. Figure 2 shows that this is uncommon. The median fix was reported by 17 stations (mean 20.1), 9 % by a single station, 85 % by three or more and 65 % by ten or more. The share of well-confirmed fixes follows the band: as high as 95 % in the daytime and as low as 32 % in the small hours.

5.3 Position: accuracy, tracks and movement

The 42 723 fixes contain 102 different six-character locators (Figure 3, Table 3). The home square KP20jg holds 86 % of the fixes and the adjacent squares (up to 5 km from it) 10 %; the square KP20kg alone, 4.6 km to the east, holds 10 %. Only 2 % of the fixes are more than 25 km from home. The farthest fix is 193 km from home, in square KP31kq on 2026-06-19. Table 4 lists the 11 trips beyond 25 km, which span 14 days. Three journeys stand out: a trip west in October 2024 (up to 143 km, 16–18 October), a trip north-east in June 2026 (up to 193 km, on 19–20 and 23 June) and a trip west in October 2026 (up to 106 km, 14–18 October, 263 fixes on the longest leg). The other trips (33–42 km, October and November 2025) are short excursions south-west and north. Home and the second most-used square are told apart only because they happen to lie in different subsquares; two places closer than about five kilometres would be indistinguishable, which is the real resolution limit of the technique.

The movement is visible in time. Between 06 and 14 UTC the home square holds 67 % of the fixes on weekdays but 88 % at weekends (Figure 4a): the weekday pattern of the car leaving the home square at

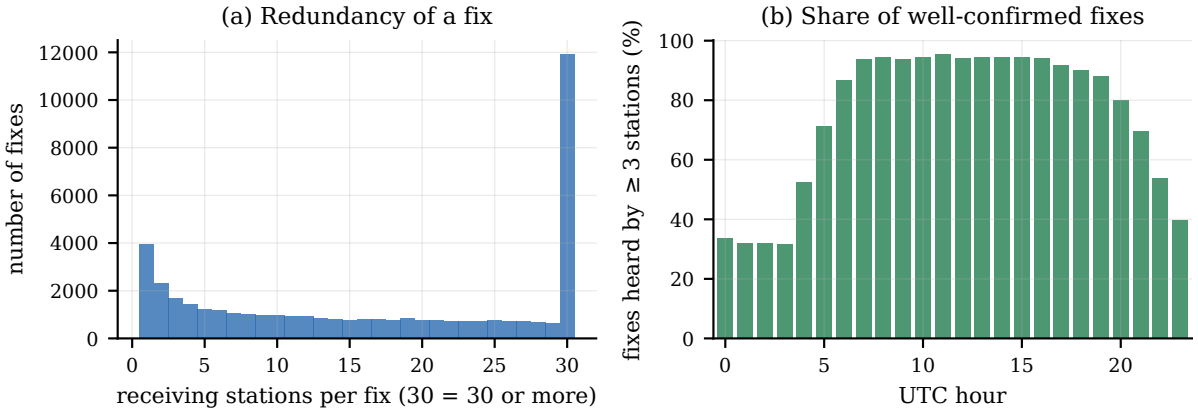


Figure 2. (a) Number of receiving stations per fix (the last bar collects 30 or more). (b) Share of fixes heard by at least three stations, by UTC hour.

the start of the working day and returning in the afternoon is recovered from the fixes alone. Between consecutive ten-minute fixes the reported square changed in only 2 % of the 38 774 steps; in the 798 cases where it did, the median jump was 4.6 km and the 90th percentile 10 km, the largest 23 km. Square-to-square jumps are quantised, so the corresponding speeds (≥62 km/h for the 90th percentile) are lower bounds; at motorway speed the car covers about 13 km in ten minutes, three squares.

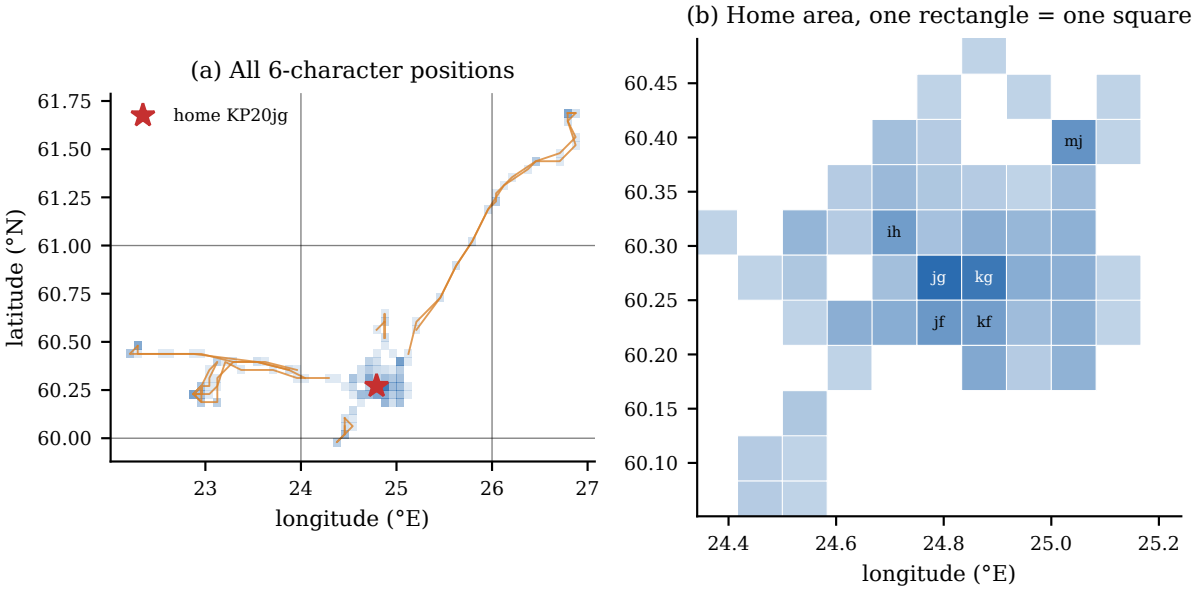


Figure 3. (a) Every six-character position recorded (shading by number of fixes; thin lines: the longest trips; black lines: 2° and 1° square boundaries). (b) The home area; each rectangle is one subsquare and the letters mark the busiest ones.

5.4 The three seasons compared

Table 5 sets the seasons side by side. The technique behaved alike each year: the median day produced 81, 90 and 96 fixes in 2024, 2025 and 2026, 81, 86 and 86 % of the fixes were confirmed by three or more stations, and the home square held 82, 89 and 83 % of them. The median number of stations per fix differs more (14, 20 and 16), following the number of active receivers and the state of the band rather than anything in the technique (see the companion study for the seasons). The seasons differ in what the car did: the farthest fix from home was 143 km in 2024, 42 km in 2025 (a year of local use) and 193 km in 2026. The 2024 season is the shortest (79 days with fixes) and 2025 and 2026 each contain about 188 and 223 days.

Table 3. Where the position fixes were: distance of the reported square centre from the home square KP20jg.

Distance from home square	Fixes	Share (%)
home square	36 556	85.6
adjacent squares (≤ 5 km)	4 479	10.5
5–25 km	962	2.3
25–100 km	82	0.2
> 100 km	644	1.5

Table 4. All trips beyond 25 km from home as seen through WPRS fixes alone (fixes less than eight hours apart are merged into one trip; consecutive days of one journey can appear as separate trips).

Start (UTC)	End (UTC)	Hours	Fixes	Squares	Max. dist. (km)
2024-10-16 11:42	10-17 19:52	32	118	6	139
2024-10-18 05:22	10-18 12:52	8	39	7	143
2025-10-12 11:22	10-12 13:22	2	13	5	40
2025-10-19 09:52	10-19 13:02	3	20	3	33
2025-11-01 07:52	11-01 08:22	0	4	3	42
2025-11-01 17:32	11-01 19:02	2	3	2	37
2026-06-19 11:12	06-20 15:52	29	165	14	193
2026-06-23 10:42	06-23 17:52	7	43	14	193
2026-10-14 10:42	10-14 18:42	8	49	8	106
2026-10-15 03:02	10-17 12:52	58	263	8	106
2026-10-18 09:02	10-18 10:22	1	9	9	106

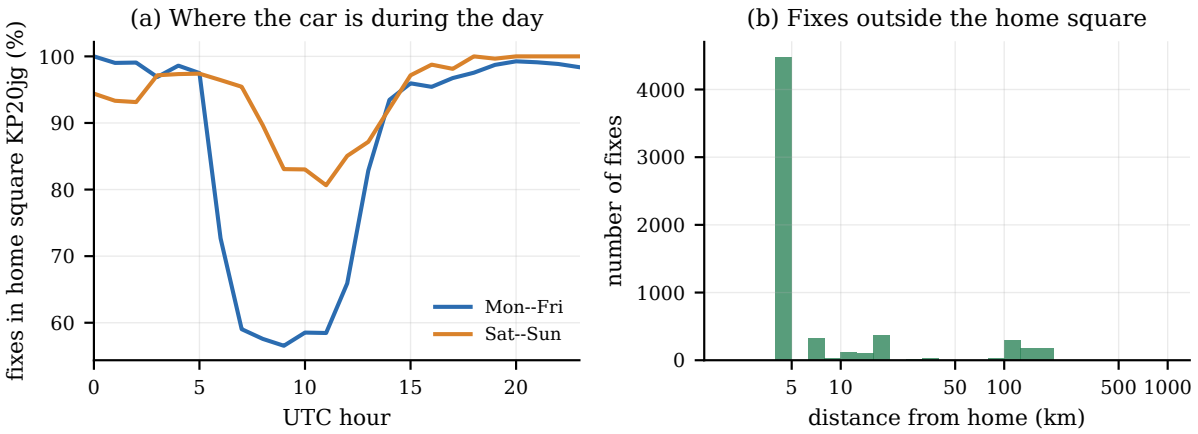


Figure 4. (a) Share of fixes in the home square by UTC hour, weekdays and weekends. (b) Distance from home of the fixes outside the home square.

Table 5. The three seasons side by side.

Quantity	2024	2025	2026
Days with fixes	79	188	223
Fixes	5 546	16 897	20 280
Median fixes per day	81	90	96
Days with ≥ 20 fixes	71	188	215
Fixes heard by ≥ 3 stations (%)	81	86	86
Median stations per fix	14	20	16
Trips beyond 25 km	2	4	5
Farthest fix from home (km)	143	42	193
Fixes in the home square (%)	82	89	83

5.5 Who hears the car: coverage and propagation

Figure 5 (taken from the companion propagation study) shows where the fixes are heard. The receiving stations are overwhelmingly European (83.5 %), most of them to the south and south-west of the transmitter, so a position fix is delivered, on a typical day, by one of the fixed receivers in Central and Western Europe, over a path of one to three thousand kilometres; 88 stations carry half of all reports. Long-distance paths add redundancy that does not matter for tracking but proves the reach: at least one station beyond 6 000 km was heard on 73 % of days, with a record of 15 619 km, i.e. 78 095 km per watt. The diurnal rhythm of availability (Figure 1b) is the rhythm of the typical day of the 20 m band (Figure 6a); the daytime plateau of availability is the plateau of reporters.

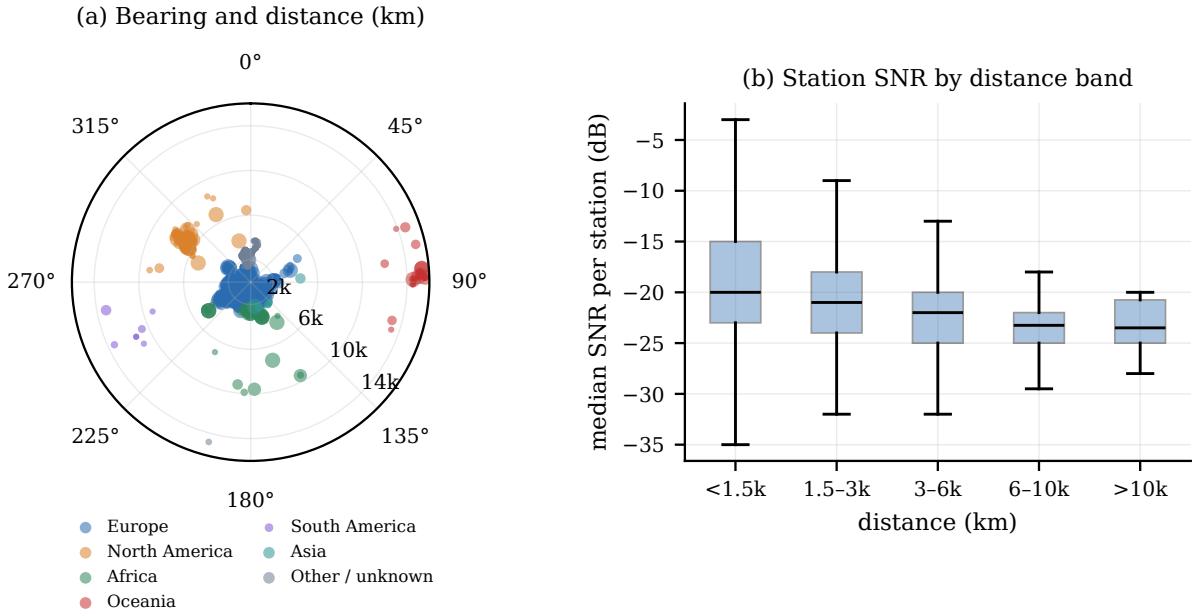


Figure 5. (Companion study.) Receiving stations by bearing and distance from the transmitter, and station SNR by distance band.

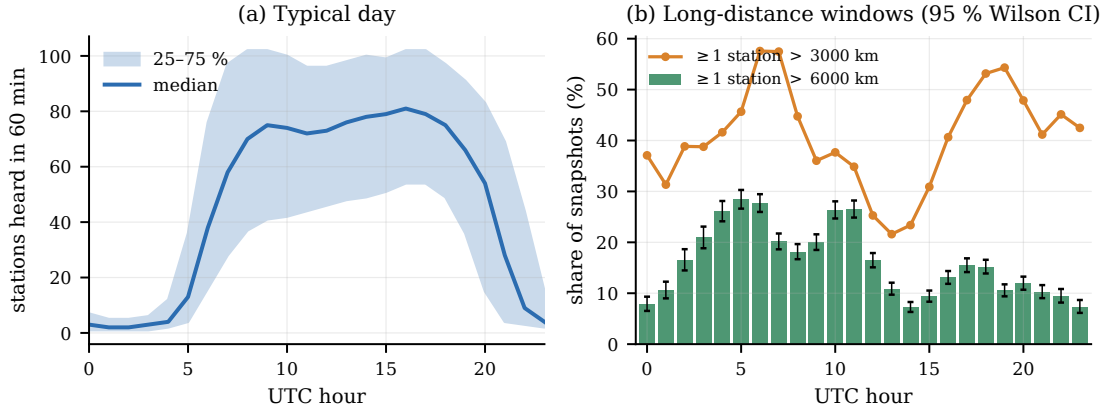


Figure 6. (Companion study.) (a) Typical day: stations heard in 60 minutes by UTC hour. (b) Share of snapshots with a station beyond 6 000 km (bars) and 3 000 km (line).

Space weather matters mainly at the margins. In the companion study the number of stations heard, adjusted for the time of day, is 27 % of the typical value when the Juliusruh MUF is below 12 MHz and 144 % of it for a MUF of 20–24 MHz; it falls modestly with the K-index and collapses on geomagnetic storm days: on the worst day of the live-logged 2026 data (4 July) only 10 stations were heard in the daytime instead of the typical 80. A fix still arrives under such conditions because a single receiver suffices, but the

updates become rarer and the confirmation weaker.

5.6 Power and reach

Table 6 and Figure 7 emulate a lower transmit power as described in Section 4. The median report is 15 dB above the decoding floor, and halving the power twice (50 mW) still leaves 98 % of the stations that heard the beacon at least once. At 25 mW 89 % still hear it, at 10 mW 70 %, at 1 mW 27 %. Stations beyond 6 000 km are much more sensitive: 86 % remain at 25 mW, 59 % at 10 mW and 10 % at 1 mW. For position reporting this is reassuring: European reception is so redundant that even 10 mW would deliver a fix to a large number of receivers; the long paths are the first to be lost. The conclusion applies to the *best* SNR ever reported by each station, and the sustained reach of a lower power would be smaller.

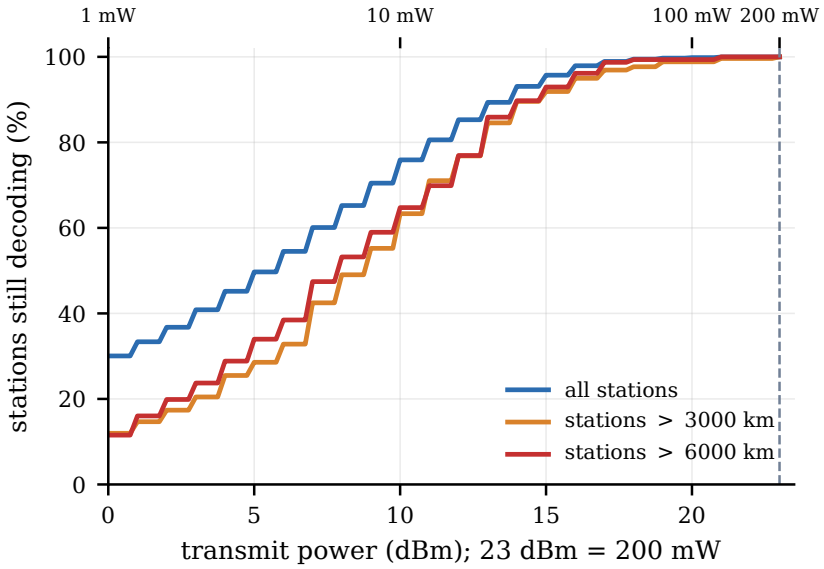


Figure 7. Emulated reach versus transmit power: share of receiving stations still above the decoding floor of −34 dB (all stations, stations beyond 3 000 km and beyond 6 000 km).

Table 6. Emulated power reduction: share of the actual reports and of the receiving stations that would still lie above the decoding floor of −34 dB if the output were lowered by Δ.

Transmit power	dBm	Δ (dB)	All reports (%)	Stations (%)	Stations > 6000 km (%)
200 mW (as used)	23	0.0	100	100	100
100 mW	20	3.0	99	100	99
50 mW	17	6.0	94	98	96
25 mW	14	9.0	81	89	86
10 mW	10	13.0	57	70	59
5 mW	7	16.0	41	55	38
1 mW	0	23.0	15	27	10

5.7 Is the car heard better away from home?

Away from home (more than 25 km, 726 fixes) the hour-adjusted number of reporters per fix is 149 % of that at home on the same days (95 % bootstrap interval 92–264 %). Only 9 days have enough fixes at both places for this comparison, so the interval is very wide and the result is inconclusive; the question cannot be answered with this dataset.

6 Discussion

6.1 WPRS compared with other position channels

Table 7 compares the characteristics at design level. The APRS and cellular entries are generic and were not measured here; the WPRS entries are the measurements of this note.

Table 7. Design-level comparison of position channels. APRS and cellular values are nominal, WPRS values are from this note.

	APRS (VHF, 144.800 MHz in Europe)	Cellular GPS tracker	WPRS (HF, 14 MHz WSPR)
Infrastructure	digipeater or internet gateway in range	mobile data coverage and a subscription	none; any one of thousands of volunteer receivers
Coverage	local, line-of-sight	where the network is	ionospheric; day and night differ
Resolution	GPS accuracy (metres)	GPS accuracy (metres)	4.6×4.6 km subsquare
Update interval	seconds to minutes	seconds to minutes	10 min, if heard
Latency	seconds	seconds	about 6–10 min
TX power	1–5 W (typical)	a fraction of a watt (cellular modem)	0.2 W
Reverse channel	messages possible	commands possible	none
Licence	amateur	none	amateur

WPRS has a niche: low cost and independence of infrastructure, at the price of coarse resolution, a ten-minute cadence, a daily gap in the small hours of the night and no way to know whether anybody listens. It is useful as an *away-from-coverage breadcrumb trail* (hiking, sailing, long drives, expeditions, a car whose owner would like to see where it was) and as a low-power propagation probe; it is not a safety tracker and cannot replace a distress beacon.

6.2 Limitations

- *One station, one car, three seasons.* All numbers are from one car and one antenna over the seasons 2024, 2025 and 2026; the data cannot show whether equipment or installation changed between them. The older seasons were rebuilt from `wspr.live` and were checked against a live-logged week (Section 4), where the rebuild differed from the stored data by a ratio of only 1.001; winter operation was not attempted.
- *No ground truth.* The true track of the car was not recorded, so the position error of the technique is the theoretical quantisation error of the locator, not a measured error against GPS logs.
- *Missing fixes are ambiguous.* A gap can be a closed band, a transmitter that lost its GPS fix, a battery swap or a car parked in a garage.
- *Public data.* The track is public. Anybody can see where the car is within about five kilometres, with ten-minute resolution, as soon as the transmitter is on; for personal security the transmitter can be switched off at home.

6.3 Possible extensions

These were not tested in this work. (i) Extra position bits could be carried in the power field of a Type 1 message (which can take only 19 distinct values) or by dedicated telemetry encodings used by balloon tracking; this would raise the resolution below the subsquare at the cost of more complicated decoding. (ii) Transmitting on a second band would add redundancy at night. (iii) The logger could show the WPRS track itself on the map, with the age of the latest fix. (iv) The schedule could be made adaptive: transmit more often when the position changes, less often when parked, which would also extend the battery life.

7 Operating recommendations

1. Switch the transmitter on and wait for the GPS lock (single LED blink ends) *before* driving off; a cold start takes longer than a warm one.
2. Put the GPS puck where it has a clear view of the sky; do not shield it with metal.
3. Make sure the magnetic mount is in contact with bare, clean roof metal and the cable is not pinched by a door.
4. Do not rely on the latest fix at night; assume the last daytime fix and the next morning.
5. For a trip longer than four days, carry a second charged power bank and swap before the first one is empty.
6. Remember that the position is public; switch off at home if that matters.

8 Conclusions

- A GPS-equipped WSPR transmitter makes a usable infrastructure-free position beacon: a six-character locator is delivered in a square of 4.6×4.6 km with an r.m.s. error of 1.9 km.
- On 20 m with 200 mW and a car-roof whip, a position from the previous hour is available 92 % of the time in the daytime and 38 % at night, and the median fix is confirmed by 17 receivers.
- The technique recovered the commute and trips to 193 km from home; its resolution limit is about five kilometres.
- A power of 25 mW would have kept 89 % of the receiving stations; the long-distance paths would have been lost first.
- A 30 000 mAh pack runs the transmitter for four to five days against a predicted 6.2.

A Reproducing the analysis

The figures, tables and numbers of this note are produced from the logger databases by `doc/src/analyze.py` (companion study and shared figures) and `doc/src/analyze_wprs.py` (this note); the paper is built with `pdflatex wprs_paper.tex` (twice). Both scripts write LaTeX macros so that no number in the text is typed by hand.

Version history

Version	Date	Changes
1.0	6 November 2026	First version: concept, station, one season of measurements.
1.1	6 November 2026	Three seasons (2024–2026): earlier seasons rebuilt from `wspr.live`; all numbers regenerated; new subsection comparing the seasons; trip list extended.
1.2	6 November 2026	Measured the effect of late reports on the rebuilt seasons: negligible (ratio 1.001).

References

[1] WSPRnet — the Weak Signal Propagation Reporter Network. `https://wsprnet.org`

[2] wspr.live — ClickHouse interface to the WSPRnet spot database. `https://wspr.live`

[3] J. Taylor, K1JT, *WSJT-X User Guide* (WSPR mode) and *WSPR 2.0 User's Guide*. `https://wsjt.sourceforge.io`

[4] ZachTek, *WSPR Desktop Transmitter* documentation. `https://www.zachtek.com/1012`; firmware and configuration software: `https://github.com/HarrydeBug/WSPR-transmitters`

[5] O. Laakso, *WSPR Logger* and companion note *Twenty-metre WSPR propagation observed from a mobile 200 mW beacon in Southern Finland*. `https://github.com/oh2gax/wspr_logger`